

ELEC 221 Lecture 09

DT Fourier series and filters (Part 2)

Thursday 8 October 2026

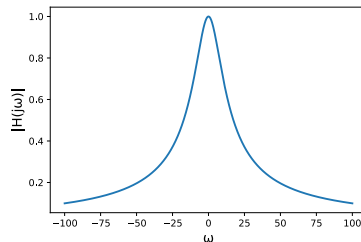
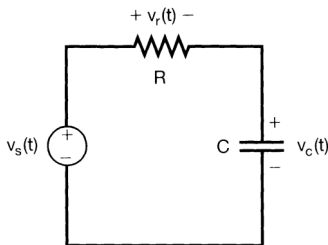
Announcements

. Tutorial assignment 2 due tomorrow

- No tutorial on Monday (Thanksgiving)
- Quiz 5 on Tuesday

Last time

We derived the frequency response for a realistic CT low pass filter (an RC circuit).



$$RC \frac{dv_c(t)}{dt} + v_c(t) = v_s(t)$$

$$v_s(t) = e^{j\omega t} \Rightarrow H(j\omega) = \frac{1}{1 + j\omega RC}$$

Last time

We explored periodic DT complex exponential signals:

$$x[n] = e^{j\omega n} = e^{j\frac{2\pi n}{N}}$$

fundamental period

We found that these signals behave differently than CT signals: we only need to consider ω in the range $[0, 2\pi)$.

$$\begin{aligned} x[n] &= e^{j(\omega + 2\pi)n} \\ &= e^{j\omega n} \cdot e^{j2\pi n} \\ &= e^{j\omega n} \end{aligned}$$

Last time

I left you with a question: what are the fundamental periods of

$$x(t) = \cos(3t), \quad \text{and} \quad x[n] = \cos(3n)$$

$$T = \frac{2\pi}{3}$$

$$N = ??$$

not periodic

Today

Learning outcomes:

- Compute the fundamental period and frequency of DT signals
- Evaluate Fourier series coefficients of DT signals
- Distinguish between finite impulse response and infinite impulse response filters in DT

Frequency and period of DT complex exponential signals

Suppose the period is N :

$$x[n+N] = x[n]$$

$$x[n] = e^{j\omega n} \Rightarrow x[n+N] = e^{j\omega(n+N)} = e^{j\omega n} e^{j\omega N} = x[n] \underbrace{e^{j\omega N}}_{\substack{\Downarrow \\ \text{must be 1}}}$$

This implies

$$e^{j\omega N} = 1 \Rightarrow \omega N = 2\pi \cdot m$$

$$m \in \mathbb{Z}$$

$$\frac{\omega}{2\pi} = \frac{m}{N}$$

$$\frac{\omega}{2\pi}$$

must be rational for the signal to be periodic.

in code: $\frac{3\pi}{23} \Rightarrow \frac{3\pi}{2\pi \cdot 23} = \frac{3}{46}$

Frequency and period of DT complex exponential signals

Exercise: what is the fundamental period of

$$x[n] = \cos(5\pi n/6) + \sin(2\pi n/3)$$

In CT: $T = \frac{2\pi}{\omega} = \frac{12}{5} \Rightarrow$ after 5 periods: at $n=12$
 $\frac{5\pi}{6} \cdot n \Rightarrow$ How big of n before have a multiple of 2π ?
 $N=12$ for $\omega/(5\pi/6)$

For $\sin(\frac{2\pi n}{3})$: $n=3$: $\frac{2\pi \cdot 3}{3} = 2\pi \checkmark \Rightarrow N=3$

$\Rightarrow$ whole signal? 12

Harmonics of DT complex exponential signals

What about harmonics?

$$x_k[n] = e^{jk \cdot \frac{2\pi}{N}n} \quad k \in \mathbb{Z}$$

In CT we had an infinite number of these. What about DT?

$$e^{j\frac{2\pi}{N}n}, e^{j\frac{2 \cdot 2\pi}{N}n}, \dots, e^{j\frac{3 \cdot 2\pi}{N}n}, \dots$$

$$e^{j \cdot N \cdot \frac{2\pi}{N}n} = e^{j \cdot 2\pi n} = 1 \Rightarrow \text{same as } k=0!$$

DT signals and LTI systems

Consider a system with impulse response $h[n]$ and DT signal $x_m[n] = e^{jm\omega n}$. Use the convolution sum:

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[n-k] h[k] \\ &= \sum_{k=-\infty}^{\infty} e^{jm\omega(n-k)} h[k] \\ &= \underbrace{e^{jm\omega n}}_{\text{CT}} \sum_{k=-\infty}^{\infty} e^{-jm\omega k} h[k] \\ &= x[n] \cdot H(e^{jm\omega}) \end{aligned}$$

$H(j\omega)$ vs. $H(e^{j\omega})$
CT vs. DT

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} e^{-j\omega k} h[k]$$

freq.
response.

DT Fourier series

complex exp. are eigenfunction in DT for LTI systems.

If we know how a system responds to complex exponential signals, we can learn its response to signals expressed in terms of them.

We need a Fourier series representation of DT signals:

$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk\omega n} \quad \omega = \frac{2\pi}{N}$$

How do we find the c_k ?

DT Fourier coefficients

$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi}{N} n}$$

Option 1: solve a system of linear equations

$$x[0] = \sum_{k=0}^{N-1} c_k = c_0 + c_1 + \dots + c_{N-1}$$

$$x[1] = \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi}{N}} = c_0 + c_1 e^{j \frac{2\pi}{N}} + \dots$$

⋮

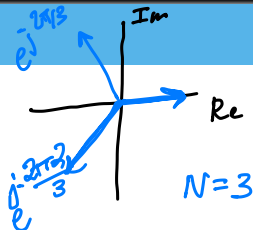
$$x[N-1] = \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi}{N} (N-1)}$$

$$\begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & e^{j \frac{2\pi}{N}} & \dots & e^{j \frac{2\pi}{N} (N-1)} \\ \vdots & \vdots & \ddots & \vdots \end{pmatrix}$$

DFT.

DT Fourier coefficients

$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi}{N} n}$$



Option 2: leverage an identity about complex numbers

$$\sum_{n=0}^{N-1} e^{jk \frac{2\pi n}{N}} = \begin{cases} N & k=0 (\pm N, \pm 2N, \dots) \\ 0 & \text{otherwise} \end{cases}$$

$$k=1: 1 + e^{j \frac{2\pi}{N}} + e^{j \frac{2\pi \cdot 2}{N}} + e^{j \frac{2\pi \cdot 3}{N}} + \dots + e^{j \frac{2\pi(N-1)}{N}} = 0$$

$$N=4$$

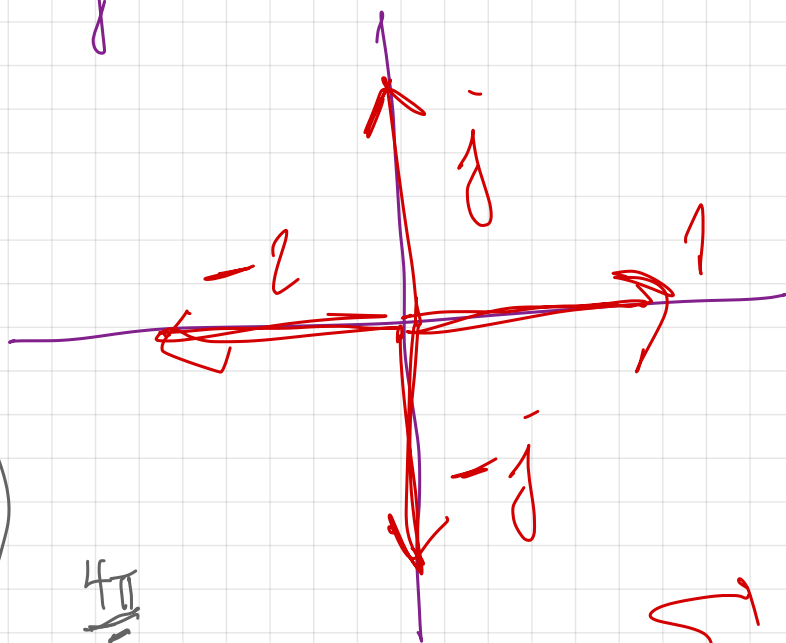
$$\sum_{n=0}^{N-1} e^{j \frac{2\pi}{4} n}$$

$$n=0$$

$$1 + e^{j \frac{2\pi}{4}} + e^{j \frac{2\pi}{4} \cdot 2} + e^{j \frac{2\pi}{4} \cdot 3}$$

$$= 1 + e^{j \pi/2} + e^{j \pi} + e^{j 3\pi/2}$$

$$\underbrace{\phantom{1 + e^{j \pi/2}}}_{j} \quad \underbrace{\phantom{e^{j \pi}}}_{-1} \quad \underbrace{\phantom{e^{j 3\pi/2}}}_{-j}$$



$$\cos\left(\frac{2\pi}{5} n\right)$$

$$\frac{4\pi}{5}$$

$$\Rightarrow \sum \rightarrow 0!$$

DT Fourier coefficients

by $e^{-jm\frac{2\pi}{N}n}$

Multiply on both sides, and sum:

$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk\frac{2\pi}{N}n}$$

$$\begin{aligned} \sum_{n=0}^{N-1} e^{-jm\frac{2\pi}{N}n} x[n] &= \sum_{n=0}^{N-1} e^{-jm\frac{2\pi}{N}n} \sum_{k=0}^{N-1} c_k e^{jk\frac{2\pi}{N}n} \\ &= \sum_{k=0}^{N-1} c_k \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}n(k-m)} \\ &= c_m \cdot N \quad \text{0 unless } m=k \end{aligned}$$

DT Fourier coefficients

DT Fourier synthesis equation

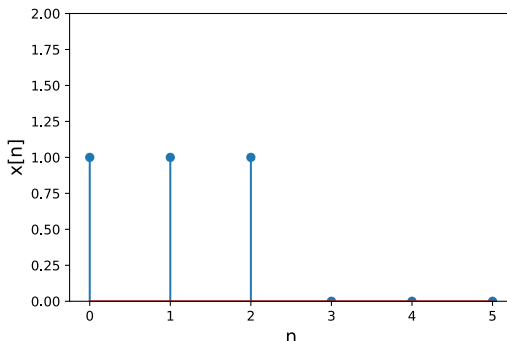
$$x[n] = \sum_{k=0}^{N-1} c_k e^{jk \frac{2\pi}{N} n}$$

DT Fourier analysis equation

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} e^{-jk \frac{2\pi}{N} n} x[n]$$

Exercise: the DT square wave

Compute the Fourier coefficients of this signal:



$$N = 6$$

$$C_k = \frac{1}{6} \sum_{n=0}^{5} x[n] e^{-jk \frac{2\pi n}{6}}$$

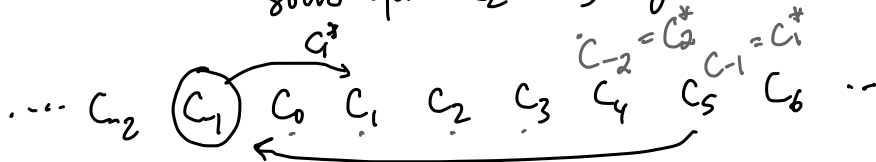
Exercise: the DT square wave

$$C_k = \frac{1}{6} \sum_{n=\langle 6 \rangle} x[n] e^{-j \frac{2\pi k n}{3}}$$

$$C_0 = \frac{1}{6} \sum_{n=\langle 6 \rangle} x[n] = \frac{1}{6} (1 + 1 + 1 + 0 + 0 + 0) = \frac{1}{2}$$

$$C_1 = \frac{1}{6} \sum_{n=\langle 6 \rangle} x[n] e^{-j \frac{2\pi n}{3}} = \frac{1}{6} (1 + e^{-j \frac{2\pi}{3}} + e^{-j \frac{4\pi}{3}}) = \frac{1}{6} (1 - \sqrt{3}j)$$

⋮
solve for $C_2 - C_5$ yourself.



$$x[n] \text{ is real} \Rightarrow C_{-k} = (C_k)^*$$

For next time

Content:

- The CT Fourier transform

Action items:

- ① Prepare for Quiz 5
- ② Tutorial Assignment 2 due Friday 23:59

Recommended reading:

- From today's class: Oppenheim 1.3.2, 1.3.3, 3.6-3.12
- Suggested problems: 1.26, 3.2, 3.10-3.12, 3.14, 3.16, 3.17, 3.19, 3.20 3.23, 3.27-3.32, 3.36, 3.39
- For next class: Oppenheim 4.1-4.3